Frontend

- Flutter (Dart) - Visitor app for Android and iOS, plus an Android camera station
- Web Interface - Admin management dashboard
- Kiosk Interface - Photo printing kiosk

Backend

- Firebase Authentication - Secure user login and account management
- Cloud Firestore (Database) - Stores visitor profiles, tickets, attractions, reservations, and photo records
- Node.js - Server-side APIs for queue reservation, voucher distribution, notification processing, system integration, and data processing
- Python - AI inference, computer vision, and email delivery invoked by the Node.js application services

Application Deployment

- Firebase Spark - Uses the existing Firebase project for authentication and shared application data
- Application Services - Node.js and Python run on the presentation computer and connect to Firebase over the internet. Background processing runs while these services are running.

Core Technology

- Live Attraction Cameras (MediaPipe EfficientDet Lite2 and WebRTC) - Android phones detect people in overlapping views and merge duplicate detections. The admin dashboard shows live video, estimated counts and Gaussian heatmaps.
- Landmark Image Recognition (OpenCV) - Matches camera images to stored landmarks to identify a starting point for navigation.
- Crowd Heatmap Analysis (Mapbox and Gaussian KDE) - Maps virtual queue demand or recent, voluntarily shared GPS positions.
- AI Recommendation System (Python) - Generates personalized attraction and itinerary recommendations.
- Image Recognition (OpenCV) - Matches visitor faces with attraction photos after opt-in selfie enrolment.
- QR Code and NFC Technology - Supports ticket validation, attraction entry and photo printing. NFC requires compatible hardware.

Objectives

-> To develop a smart theme park management system that enhances visitor experience through digital services and intelligent technologies.
-> To implement AI-based recommendation and navigation features that help visitors locate attractions, facilities, and food options efficiently.
-> To monitor attraction crowds using live cameras, estimated counts and heatmaps.
-> To implement a virtual queue reservation system that reduces waiting time and improves crowd management.
-> To provide automated photo recognition services that allow visitors to easily access and purchase attraction photos.
-> To support theme park management with crowd monitoring, facility management, and data analytics to improve operational efficiency.

Business Value

1. Improve Visitor Experience
Visitors can easily navigate the theme park, book tickets, reserve rides, and receive personalized recommendations through a mobile application.
2. Increase Operational Efficiency
Virtual queue reservation allows visitors to reserve rides digitally, minimizing long physical queues.
3. Better Crowd Management
Live attraction cameras and crowd maps help staff spot busy areas and decide where to adjust operations.
4. Increased Revenue Opportunities
Features such as photo purchases, vouchers, and personalized promotions increase visitor spending within the park.
5. Operational Efficiency
Digital ticketing, automated customer support, and centralized management tools help reduce manual work and improve operational efficiency.

Visitor

User Authentication Module
- Register and log in using email and password
- Unlock a retained session with Face ID / fingerprint on supported devices
- Secure password management
- Recover access and reset the password using an email one-time code

User Profile Management Module
- Manage personal profile information (eg. Name, Height, Food Preferences, Interests, etc.)
- Upload selfie for facial recognition registration (optional, used for photo recognition feature)
- Reset password
- Delete account
- Submit feedback or complaints

- View tutorials and user guides
- Scan QR code to add purchased tickets
- View current and past visited attractions
- View loyalty points and redemption history
- View promotion vouchers issued by the system

Smart Navigation & Map Module
- Display theme park map and navigation routes
- View attraction details including availability status and ratings
- Show outdoor GPS location when the visitor grants permission
- Identify a starting point by matching camera images to known landmarks
- Multilingual AI chatbot for visitor assistance
- Get walking directions to attractions and facilities

Ticket Purchase Module
- Book theme park tickets for individuals or groups in supported currencies
- Apply loyalty points and confirm bookings without a bank or card charge
- Generate digital tickets with QR and NFC on supported devices
- View ticket purchase history

Group Management Module
- CRUD visitor groups
- Add or remove group members
- Automatically link tickets for app users
- Manually link tickets via QR code
- View live GPS locations shared by group members who opt in
- Make group attraction reservations

Virtual Queue Reservation Module
- Analyze crowd density using heatmap technology
- Generate itinerary recommendations based on distance, crowd level, and user preferences
- Reserve attraction time slots digitally
- Support group reservations
- Opt in to reservation reminders via native notifications or supported browser Web Push
- Apply penalty for repeated no-shows (configurable by admin)
- Generate digital attraction entrance (QR or NFC)
- Automatically collect loyalty points based on promotions
- Submit attraction ratings and feedback

Smart Photo Recognition Module
- Automatically match visitor faces with attraction photos uploaded by photographers
- View matched photos in preview mode
- Apply photo editing features (filters, stickers, beauty effects)
- Purchase digital photos and receive softcopy via email

- Send photos for printing at photo printing kiosk
- Generate QR or NFC for kiosk printing
- View photo purchase history

Photo Printing Kiosk

Photo Printing Kiosk module
- Scan QR or NFC to retrieve photo printing requests
- Preview and edit photos before printing
- Apply filters, stickers, or photo enhancements
- Confirm photo orders and award loyalty points where applicable

Admin

Visitor Management Module
- View visitor accounts and profiles
- Monitor visitor activities
- Detect suspicious accounts based on abnormal login, reservation, or payment behavior
- Flag repeated unusual activity for staff review
- Suspend or deactivate suspicious accounts
- Manage visitor feedback and complaints

Ticket Management Module
- Manage ticket pricing and ticket packages
- Configure ticket purchase rules
- Manage promotions and discount campaigns

Attraction & Facility Management Module
- CRUD attraction information
- Update ride status (Open / Closed / Maintenance)
- CRUD facility details such as restaurants, shops, and rest areas
- Update attraction information displayed on the park map

Crowd Monitoring & Analytics Module
- View live video, estimated people counts and heatmaps for assigned attraction cameras
- Review virtual queue demand and identify busy attractions
- Analyze peak check-in hours and aggregated attraction journeys
- Open facility controls to update attraction availability

Photo Management Module
- Upload attraction photos taken by photographers
- Manage photo recognition system records
- Monitor photo purchase transactions

Voucher & Promotion Management Module
- Create and manage promotional vouchers
- Issue rain compensation vouchers automatically
- Track voucher usage and redemption rates
- Monitor weather conditions affecting park operations Trigger compensation vouchers if required conditions are met

Loyalty & Rewards Management Module
- Configure loyalty point collection rules
- Manage reward redemption items
- Monitor loyalty point transactions

Reporting & Analytics Module
- Generate analytical reports (eg. Top 10 popular attractions, Sales summary reports, Visitor statistics)
- Display reports using charts and graphs
- Export reports in PDF format

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The system integrates advanced technologies such as AI recommendation systems, facial recognition, and smart navigation to enhance theme park infrastructure and visitor services.

Implementation and Presentation Verification

- Verified live video, people counts and heatmaps using the OnePlus 11 and PC browser.
- Confirmed the TAR UMT Penang map and email delivery for password recovery and photo orders.
- Rehearse the full flow on the presentation PC and Wi-Fi, including multiple feeds if used.
- Verify physical NFC, printer output and closed-app reminders on the target devices.
- Verify the iOS build, and test rain voucher issuance and redemption after qualifying weather.